



Lecture 7: Multiple continuous random variables

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April 6, 2021

1. Review
2. Multiple continuous random variables
3. Conditioning and independence
4. The continuous Bayes' rule
5. Recitation
6. Summary



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- Summary of concepts

$$p_X(x)$$

$$f_X(x)$$

$$F_X(x)$$

$$\sum_x x p_X(x)$$

$$\mathbf{E}[X]$$

$$\int x f_X(x) dx$$

$$\text{var}(X)$$

$$p_{X,Y}(x, y)$$

$$f_{X,Y}(x, y)$$

$$p_{X|A}(x)$$

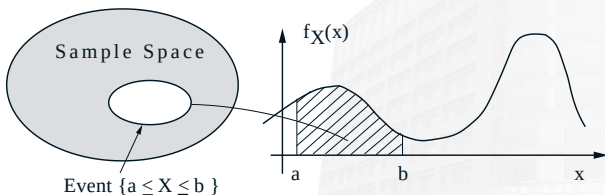
$$f_{X|A}(x)$$

$$p_{X|Y}(x | y)$$

$$f_{X|Y}(x | y)$$

Continuous RVs

- $\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$
- $\mathbf{P}(x \leq X \leq x + \delta) \approx f_X(x) \cdot \delta$
- $\mathbf{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$



- Cumulative distribution functions

$$F_X(x) = \mathbf{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

- Normal RVs

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

- Exponential RVs

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

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Definition of joint PDFs

The two continuous RVs X and Y , with the same experiment, are **jointly continuous** if they can be described by a **joint PDF** $f_{X,Y}$, where $f_{X,Y}$ is a nonnegative function that satisfies

$$\mathbf{P}((X, Y) \in B) = \iint_{(x,y) \in B} f_{X,Y}(x, y) dx dy$$

for every subset B of the two-dimensional plane. In particular, when B is of the form $B = \{(x, y) | a \leq x \leq b, c \leq y \leq d\}$, we have

$$\mathbf{P}(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f_{X,Y}(x, y) dx dy.$$

$$\mathbf{P}((X, Y) \in B) = \iint_{(x,y) \in B} f_{X,Y}(x, y) dx dy$$

- Normalization:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1.$$

- Interpretation (small rectangle):

$$\mathbf{P}(a \leq X \leq a + \delta, c \leq Y \leq c + \delta) \approx f_{X,Y}(a, c) \cdot \delta^2$$

- Marginal PDF:

$$\mathbf{P}(X \in A) = \mathbf{P}(X \in A, Y \in (-\infty, \infty)) = \int_A \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

Definition of joint CDFs

If X and Y are two RVs associated with the same experiment, we define their **joint CDF** by

$$F_{X,Y}(x, y) = \mathbf{P}(X \leq x, Y \leq y).$$

- If X, Y are described by a joint PDF $f_{X,Y}$,

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(s, t) ds dt$$

- Conversely,

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}(x, y).$$

- Expectations:

$$\mathbf{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy$$

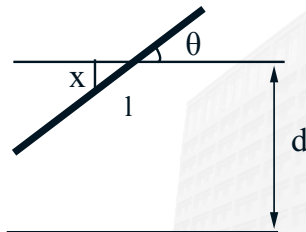
If g is linear, of the form $aX + bY + c$,

$$\mathbf{E}[aX + bY + c] = a\mathbf{E}[X] + b\mathbf{E}[Y] + c$$

- More than two RVs, e.g., $f_{X,Y,Z}$
- X and Y are called **independent** if

$$f_{X,Y}(x, y) = f_X(x) f_Y(y), \quad \text{for all } x, y$$

- Parallel lines are at distance d , throw a needle of length ℓ (assume $\ell < d$)
- Find $\mathbf{P}$ (needle intersects one of the lines)
- $X \in [0, d/2]$: distance of needle midpoint to the nearest line



- **Model:** X, Θ uniform, independent
- $f_{X,\Theta}(x, \theta) = \frac{2}{d} \cdot \frac{2}{\pi} \quad (0 \leq x \leq d/2, 0 \leq \theta \leq \pi/2)$
- Intersect if $X \leq \frac{\ell}{2} \sin \Theta$



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- Intersect if $X \leq \frac{\ell}{2} \sin \Theta$

$$\begin{aligned} \mathbf{P} \left(X \leq \frac{\ell}{2} \sin \Theta \right) &= \iint_{x \leq \frac{\ell}{2} \sin \theta} f_X(x) f_\Theta(\theta) dx d\theta \\ &= \frac{4}{\pi d} \int_0^{\pi/2} \int_0^{(\ell/2) \sin \theta} dx d\theta \\ &= \frac{4}{\pi d} \int_0^{\pi/2} \frac{\ell}{2} \sin \theta d\theta = \frac{2\ell}{\pi d} \end{aligned}$$

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Definition of Conditional PDFs

Let X and Y be continuous RVs with joint PDF $f_{X,Y}$. For any $f_Y(y) > 0$, the **conditional PDF** of X given $Y = y$ is defined by

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

- Discrete case: $p_{X|Y}(x | y) = p_{X,Y}(x, y)/p_Y(y)$
- Recall

$$\mathbf{P}(x \leq X \leq x + \delta) \approx f_X(x) \cdot \delta$$

- By analogy, **for fixed y** , would like:

$$\mathbf{P}(x \leq X \leq x + \delta | Y = y) \approx f_{X|Y}(x | y) \cdot \delta$$

- Technical difficulty: $\{Y = y\}$ is a zero-probability event in

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- Let event $B = \{y \leq Y \leq y + \epsilon\}$, for small ϵ and consider the **probability** instead of PDF:

$$\begin{aligned} \mathbf{P}(x \leq X \leq x + \delta \mid B) &= \frac{\mathbf{P}(x \leq X \leq x + \delta, y \leq Y \leq y + \epsilon)}{\mathbf{P}(y \leq Y \leq y + \epsilon)} \\ &\approx \frac{f_{X,Y}(x, y) \delta \epsilon}{f_Y(y) \epsilon} = \frac{f_{X,Y}(x, y)}{f_Y(y)} \cdot \delta \end{aligned}$$

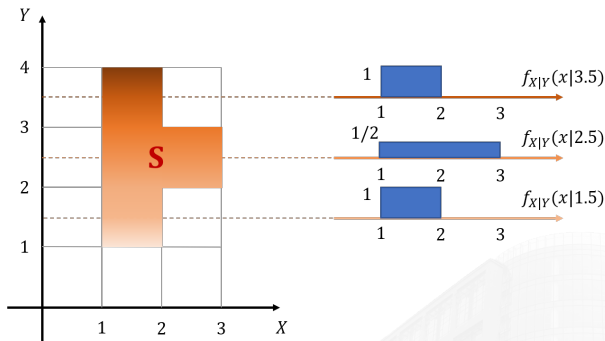
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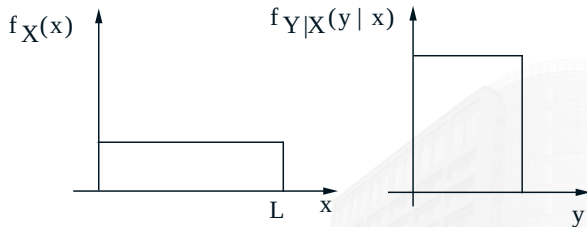
- Limiting case when $\epsilon \rightarrow 0$, to define conditional PDF where the denominator is a zero-probability event.



- Let X and Y have a joint PDF which is uniform on the set S .
- For each fixed y , we consider the joint PDF along the slice $Y = y$ and normalize it so that it integrates to 1.

Stick-breaking example

- Break a stick of length ℓ twice: break at X : uniform in $[0, \ell]$; break again at Y , uniform in $[0, X]$
- Find $\mathbf{E}[Y]$



$$f_{X,Y}(x, y) = f_X(x)f_{Y|X}(y | x) = \frac{1}{\ell} \cdot \frac{1}{x}$$

- Joint PDF:

$$f_{X,Y}(x,y) = \frac{1}{\ell} \cdot \frac{1}{x}, \quad 0 \leq y \leq x \leq \ell$$

- Marginal PDF:

$$\begin{aligned} f_Y(y) &= \int f_{X,Y}(x,y) dx \\ &= \int_y^\ell \frac{1}{\ell x} dx \\ &= \frac{1}{\ell} \ln \frac{\ell}{y}, \quad 0 \leq y \leq \ell \end{aligned}$$

- Joint PDF:

$$f_{X,Y}(x,y) = \frac{1}{\ell} \cdot \frac{1}{x}, \quad 0 \leq y \leq x \leq \ell$$

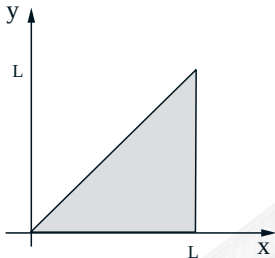
- Marginal PDF:

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- Expectation:

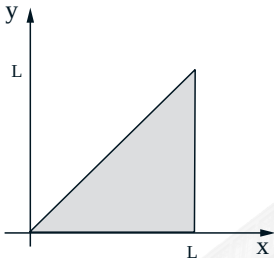
$$\mathbf{E}[Y] = \int_0^\ell y f_Y(y) dy = \int_0^\ell y \frac{1}{\ell} \ln \frac{\ell}{y} dy = \frac{\ell}{4}$$

- Alternative: Conditional expectation



$$\mathbf{E}[Y \mid X = x] = \int_0^x y f_{Y|X}(y \mid x) dy = \frac{x}{2}$$

- Alternative: Conditional expectation



$$\mathbf{E}[Y \mid X = x] = \int_0^x y f_{Y|X}(y \mid x) dy = \frac{x}{2}$$

$$\mathbf{E}[Y] = \int_0^\ell \mathbf{E}[Y \mid X = x] f_X(x) dx = \frac{\ell}{4}$$

Definition of Conditional Expectations

The conditional expectation of X given that A has happened is defined by

$$\mathbf{E}[X \mid A] = \int_{-\infty}^{\infty} x f_{X|A}(x) dx.$$

For a function $g(X)$, we have

$$\mathbf{E}[g(X) \mid A] = \int_{-\infty}^{\infty} g(x) f_{X|A}(x) dx.$$

- **Total expectation theorem:** Let $A_1, A_2, \dots, A_n$ be disjoint events that form a partition of the sample space, and $\mathbf{P}(A_i) > 0$ for all i . Then,

$$\mathbf{E}[X] = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{E}[X \mid A_i].$$

Definition of Conditional Expectations

The conditional expectation of X given that $Y = y$ has happened is defined by

$$\mathbf{E}[X \mid Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx.$$

For a function $g(X)$, we have

$$\mathbf{E}[g(X) \mid Y = y] = \int_{-\infty}^{\infty} g(x) f_{X|Y}(x|y) dx.$$

- **Total expectation theorem:**

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} \mathbf{E}[X \mid Y = y] f_Y(y) dy.$$

- Suppose that the RV X has the piecewise constant PDF

$$f_X(x) = \begin{cases} 1/3, & \text{if } 0 \leq x \leq 1, \\ 2/3, & \text{if } 1 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

- Consider the events

$$A_1 = \{X \text{ lies in the first interval } [0, 1]\}$$

$$A_2 = \{X \text{ lies in the second interval } (1, 2]\}$$

- We have from the given PDF

$$\mathbf{P}(A_1) = \int_0^1 f_X(x)dx = \frac{1}{3}, \quad \mathbf{P}(A_2) = \int_1^2 f_X(x)dx = \frac{2}{3}$$

- Recall that the mean of a uniform RV over an interval $[a, b]$ is $(a + b)/2$ and its second moment is $(a^2 + ab + b^2)/3$.

$$\mathbf{E}[X \mid A_1] = \frac{1}{2}, \quad \mathbf{E}[X \mid A_2] = \frac{3}{2}$$

$$\mathbf{E}[X^2 \mid A_1] = \frac{1}{3}, \quad \mathbf{E}[X^2 \mid A_2] = \frac{7}{3}$$

- By the total expectation theorem,

$$\mathbf{E}[X] = \mathbf{P}(A_1)\mathbf{E}[X \mid A_1] + \mathbf{P}(A_2)\mathbf{E}[X \mid A_2] = \frac{7}{6}$$

$$\mathbf{E}[X^2] = \mathbf{P}(A_1)\mathbf{E}[X^2 \mid A_1] + \mathbf{P}(A_2)\mathbf{E}[X^2 \mid A_2] = \frac{15}{9}$$

- The variance is given by

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \frac{11}{36}$$

- Two continuous RVs X and Y are **independent** if

$$f_{X,Y}(x,y) = f_X(x)f_Y(y), \quad \text{for all } x, y.$$

- Independence is the same as the condition

$$f_{X|Y}(x | y) = f_X(x), \quad \text{for all } y \text{ with } f_Y(y) > 0 \text{ and all } x.$$

- If X and Y are independent, then

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

$$\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[h(Y)] \quad \forall g, h$$

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

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The continuous Bayes' rule

- From the formulas $f_{X,Y} = f_X f_{Y|X} = f_Y f_{X|Y}$, it follows that

$$f_{X|Y}(x | y) = \frac{f_X(x) f_{Y|X}(y | x)}{f_Y(y)}$$

- Based on the normalization property $\int_{-\infty}^{\infty} f_{X|Y}(x | y) dx = 1$,

$$f_{X|Y}(x | y) = \frac{f_X(x) f_{Y|X}(y | x)}{\int_{-\infty}^{\infty} f_X(t) f_{Y|X}(y | t) dt}.$$

- A light bulb is known to have an exponentially distributed lifetime Y . The parameter λ of the pdf is uniformly distributed in the interval $[1, 3/2]$. We test a light bulb and record its lifetime. What can we say about the underlying λ ?

- A light bulb is known to have an exponentially distributed lifetime Y . The parameter λ of the pdf is uniformly distributed in the interval $[1, 3/2]$. We test a light bulb and record its lifetime. What can we say about the underlying λ ?

Solution:

- We model λ in terms of a uniform RV Λ with PDF

$$f_{\Lambda}(\lambda) = 2, \quad \text{for } 1 \leq \lambda \leq \frac{3}{2}.$$

$$f_{Y|\Lambda}(y | \lambda) = \lambda e^{-\lambda y}$$

- The available information about Λ is captured by the conditional pdf $f_{\Lambda|Y}(\lambda | y)$

$$f_{\Lambda|Y}(\lambda | y) = \frac{f_{\Lambda}(\lambda) f_{Y|\Lambda}(y | \lambda)}{\int_{-\infty}^{\infty} f_{\Lambda}(t) f_{Y|\Lambda}(y | t) dt}, \quad \text{for } 1 \leq \lambda \leq \frac{3}{2}.$$

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1. Let X be an exponential random variable with parameter $\lambda > 0$. Calculate the probability that X belongs to one of the intervals $[n, n + 1]$ with n odd.

2. **Exponential Random Variable is Memoryless.** The time T until a new light bulb burns out is an exponential random variable with parameter λ . Ariadne turns the light on, leaves the room, and when she returns, t time units later, finds that the bulb is still on, which corresponds to the event $A = \{T > t\}$. Let X be the additional time until the bulb burns out. What is the conditional CDF of X , given the event A ?

3. Let the random variables X and Y have a joint pdf which is uniform over the triangle with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$.

- (a) Find the joint pdf of X and Y .
- (b) Find the marginal pdf of Y .
- (c) Find the conditional pdf of X given Y .
- (d) Find $\mathbf{E}[X|Y = y]$, and use the total expectation theorem to find $\mathbf{E}[X]$ in terms of $\mathbf{E}[Y]$.
- (e) Use the symmetry of the problem to find the value of $\mathbf{E}[X]$.

4. We have a stick of unit length, and we break it into three pieces. We choose randomly and independently two points on the stick using a uniform pdf, and we break the stick at these points. What is the probability that the three pieces we are left with can form a triangle?

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- Joint pdf

$$\mathbf{P}((X, Y) \in S) = \iint_S f_{X,Y}(x, y) dx dy$$

- Conditioning

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \quad \text{if } f_Y(y) > 0$$

- The continuous Bayes' rule

$$f_{X|Y}(x | y) = \frac{f_X(x)f_{Y|X}(y | x)}{\int_{-\infty}^{\infty} f_X(t)f_{Y|X}(y | t) dt}$$

- Readings: Section 3.4-3.6
- Assignment: Problem Set 7
- Next lecture: Derived distributions, convolution